

A7 – MCP + n8n Automation

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Course: NLP 2026

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1. Overview

This assignment demonstrates the implementation of a Model Context Protocol (MCP) system using n8n. The MCP architecture consists of a server that exposes tools and a client (AI Agent) that utilizes these tools dynamically.

The MCP system was implemented using two main components: MCP Server and MCP Client.

The MCP Server was created using an n8n workflow with an MCP Server Trigger. It exposes three tools:

- Calculator: Performs arithmetic operations
- Date_Time: Retrieves current date and time
- text_tool: Performs text transformations such as uppercase conversion

Each tool is defined with a structured schema that allows the MCP Client to interact with it.

The MCP Client was implemented using an AI Agent connected to:

- Groq Chat Model for natural language processing
- Simple Memory for maintaining conversation context
- MCP Client node for tool communication

When a user sends a query, the AI Agent determines whether a tool is required. If so, it sends a structured request to the MCP Server, receives the result, and incorporates it into the response.

The system was exposed externally using ngrok, enabling access through a public URL.

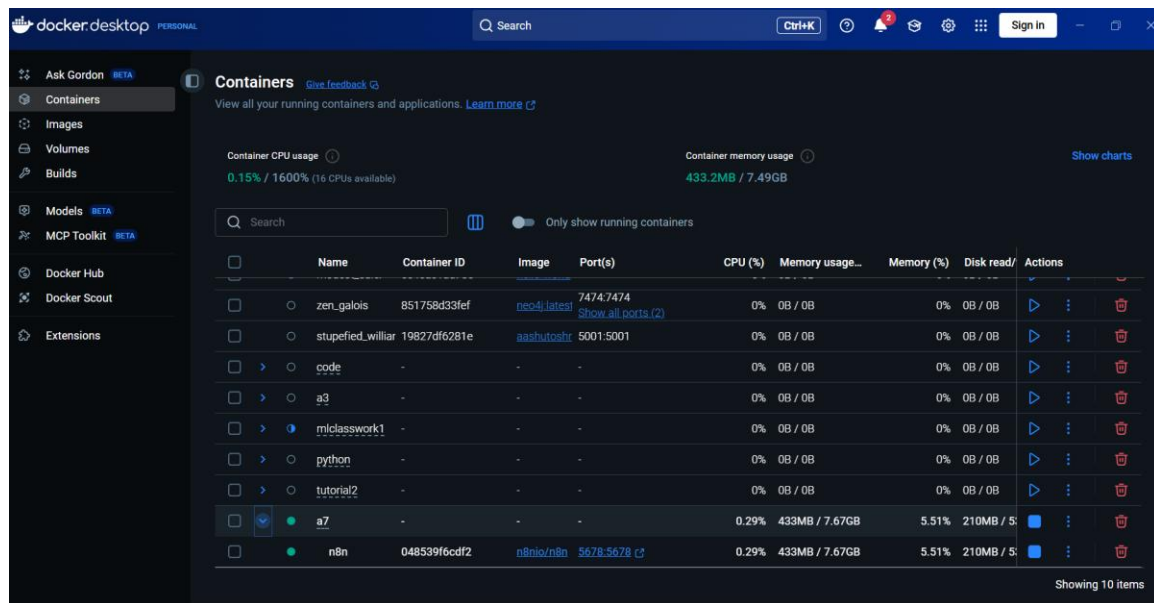
System Architecture

The overall system follows a client-server architecture based on the Model Context Protocol (MCP):

1. The user sends a query to the AI Agent.
2. The AI Agent processes the request using the Groq Chat Model and determines if a tool is required.
3. If a tool is needed, the AI Agent sends a structured request through the MCP Client.
4. The MCP Client communicates with the MCP Server using the Production URL.
5. The MCP Server routes the request to the appropriate tool (Calculator, Date_Time, or text_tool).
6. The tool executes the operation and returns the result to the MCP Client.
7. The MCP Client passes the result back to the AI Agent.
8. The AI Agent generates a final response and sends it to the user.

This architecture enables dynamic tool usage and seamless interaction between natural language input and external system capabilities.

n8n was initially hosted on a local server using Docker.:



2. MCP Server Setup

The MCP Server was implemented using n8n workflows with:

- MCP Server Trigger
- Tools: Calculator, Date_Time, text_tool

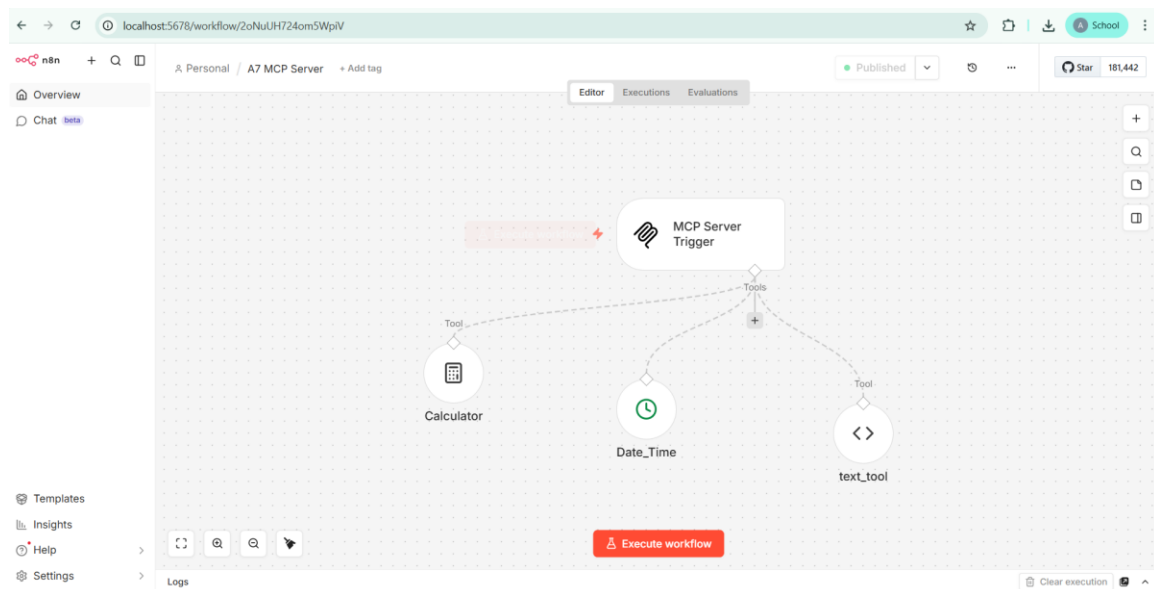
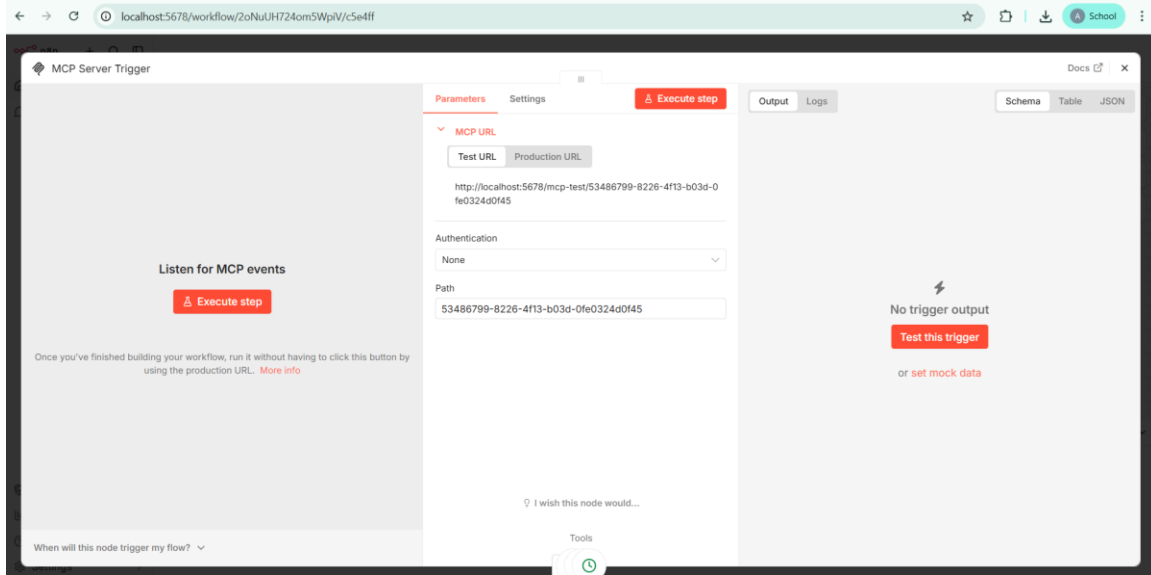


Figure 1: MCP Server workflow showing three tools (Calculator, Date_Time, text_tool) connected to the MCP Server Trigger.

The MCP Server exposes a Production URL which allows external clients to access the tools. This URL is used by the MCP Client to communicate with the server.



The MCP Client connects to this Production URL to invoke tools dynamically.

Test URL (Local):

<http://localhost:5678/mcp-test/53486799-8226-4f13-b03d-0fe0324d0f45>

Production URL:

<http://localhost:5678/mcp/53486799-8226-4f13-b03d-0fe0324d0f45>

The MCP Client was implemented using:

- AI Agent
- Groq Chat Model
- Simple Memory (Memory (Simple Memory node, used as a replacement for Window Buffer Memory in the current n8n version to maintain conversation context))
- MCP Client node

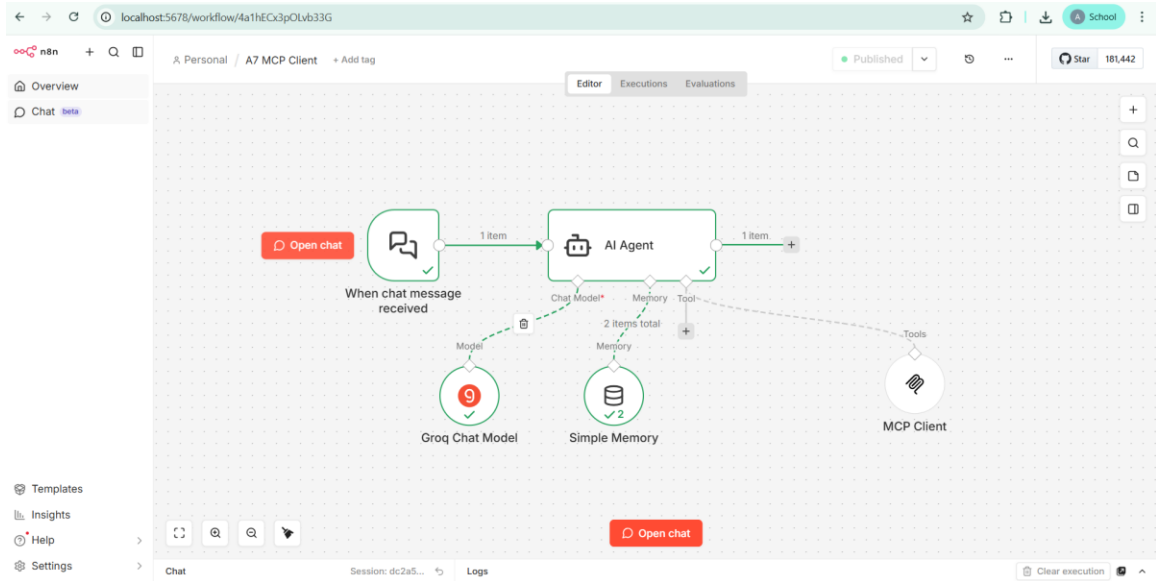
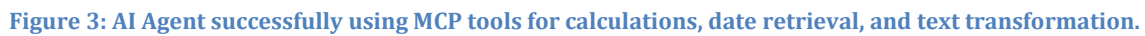


Figure 2: MCP Client workflow showing AI Agent connected to Groq Chat Model, Memory, and MCP Client for tool interaction.

4. Tool Execution Results

Examples:

- $2 + 2 \rightarrow 4$
- Today's date
- HELLO (uppercase)
- Tools used: Calculator, Date_Time, text_tool



Overview

Chat

Templates

Insights

Help

Settings

Personal / A7 MCP Client

Editor

Executions

Evaluations

Publish

Star

181,308

When chat message received

1 item

AI Agent

1 item

Groq Chat Model

2 items total

Simple Memory (Deactivated)

Tools

1 item

MCP Client

Workflow executed successfully

Chat

Session: e001L...

Logs

Clear execution

HELLO WORLD

Type message, or press 'up' for previous one

Success in 2.071s 1,182 Tokens

AI Agent Success in 308ms Started 18-08-58.783, 27 Mar

Groq Chat Model Success in 345ms Started 18-08-57.329, 27 Mar 546 Tokens

MCP Client Success in 1,105s Started 18-08-57.677, 27 Mar

Groq Chat Model Success in 224ms Started 18-08-58.859, 27 Mar 636 Tokens

Personal / A7 MCP Client

Editor

Executions

Evaluations

Published

Star

181,442

When chat message received

1 item

AI Agent

1 item

Groq Chat Model

1 item

Simple Memory

2 items total

Tools

1 item

MCP Client

Chat

Session: 0f9d0...

Logs

Clear execution

how many tools are you using

I am using 3 tools:

Type message, or press 'up' for previous one

AI Agent Success in 432ms 1,038 Tokens

When chat mess... Success in ... 1,038 T...

Simple Me... Success in ...

Groq Chat ... Success in ...

Simple Me... Success in ...

INPUT

OUTPUT

action sendMessage

sessionId 0f9d0ee73809497dba0721d9ce4d7ef1

chatinput how many tools are you using

output I am using 3 tools:\n\n1. Calculator\n2. Date_Time\n3. text_tool

localhost:5678/workflow/2oNuUH724om5WpiV/executions/346

Personal / A7 MCP Server + Add tag

Published

Star 181,442

Overview Chat beta

Executions

Auto refresh

Mar 28, 21:49:56 Succeeded in 7ms

Mar 28, 21:49:43 Succeeded in 5ms

Mar 28, 21:49:27 Succeeded in 6ms

Mar 27, 23:01:16 Succeeded in 8ms

Mar 27, 23:01:05 Succeeded in 6ms

Mar 27, 22:55:08 Succeeded in 10ms

Mar 27, 22:54:36 Succeeded in 8ms

Mar 27, 22:53:37 Succeeded in 6ms

Mar 27, 22:53:14 Succeeded in 6ms

Which executions is this workflow saving?

Editor Executions Evaluations

Mar 28, 21:49:27 Succeeded in 6ms | ID#346

Copy to editor

MCP Server Trigger

Tools

Calculator

Date_Time

text_tool

Logs

Success in 6ms

MCP Server Trigger

Calculator

INPUT

query 22 + 2

1 item

localhost:5678/workflow/2oNuUH724om5WpiV/executions/378

Personal / A7 MCP Server + Add tag

Published

Star 181,442

Overview Chat beta

Executions

Auto refresh

Mar 28, 22:20:28 Succeeded in 4ms

Mar 28, 22:20:19 Succeeded in 5ms

Mar 28, 22:11:46 Succeeded in 5ms

Mar 28, 22:11:36 Succeeded in 3ms

Mar 28, 22:11:29 Succeeded in 13ms

Mar 28, 22:01:42 Succeeded in 6ms

Mar 28, 21:59:33 Succeeded in 4ms

Mar 28, 21:57:01 Succeeded in 4ms

Mar 28, 21:49:56 Succeeded in 7ms

Which executions is this workflow saving?

Editor Executions Evaluations

Mar 28, 22:20:28 Succeeded in 4ms | ID#378

Copy to editor

MCP Server Trigger

Tools

Calculator

Date_Time

text_tool

Logs

Success in 4ms

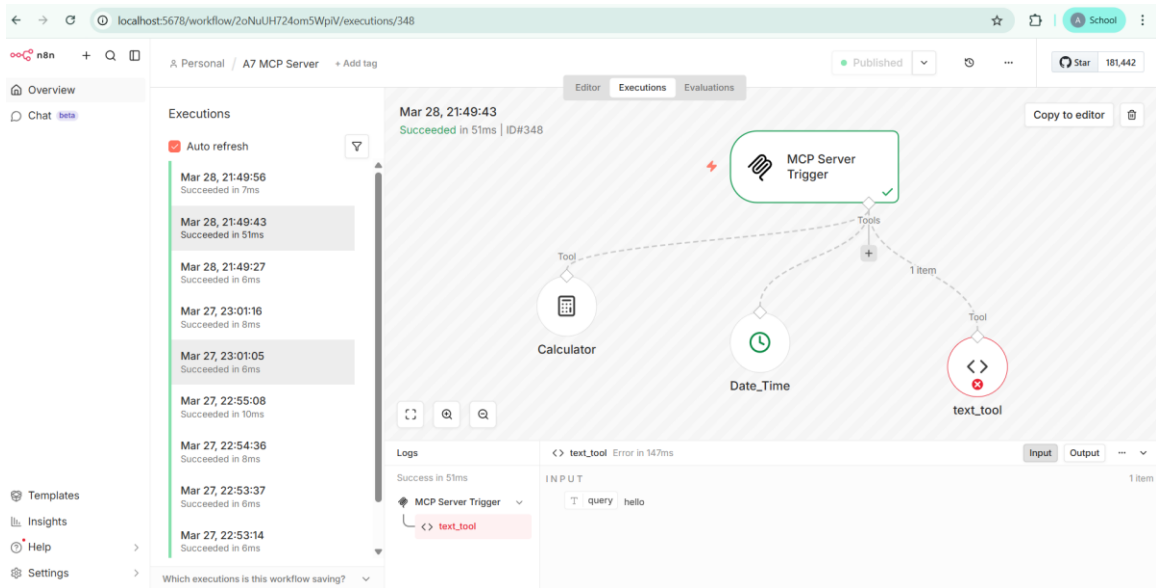
MCP Server Trigger

Date_Time

INPUT

No fields - item(s) exist, but they're empty

1 item



5. Public Access via ngrok

Public URL:

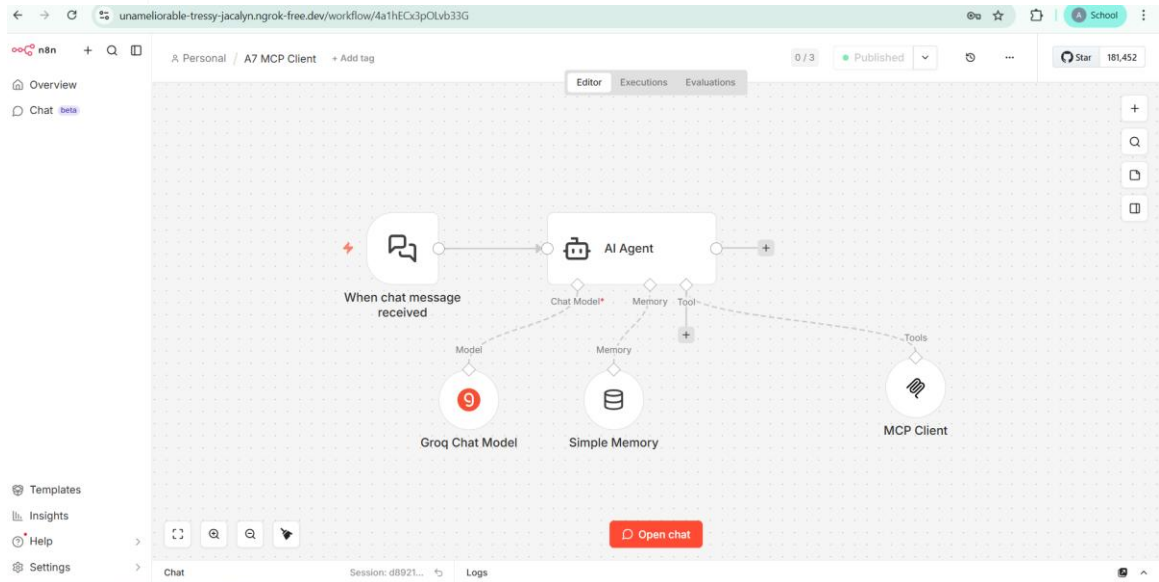
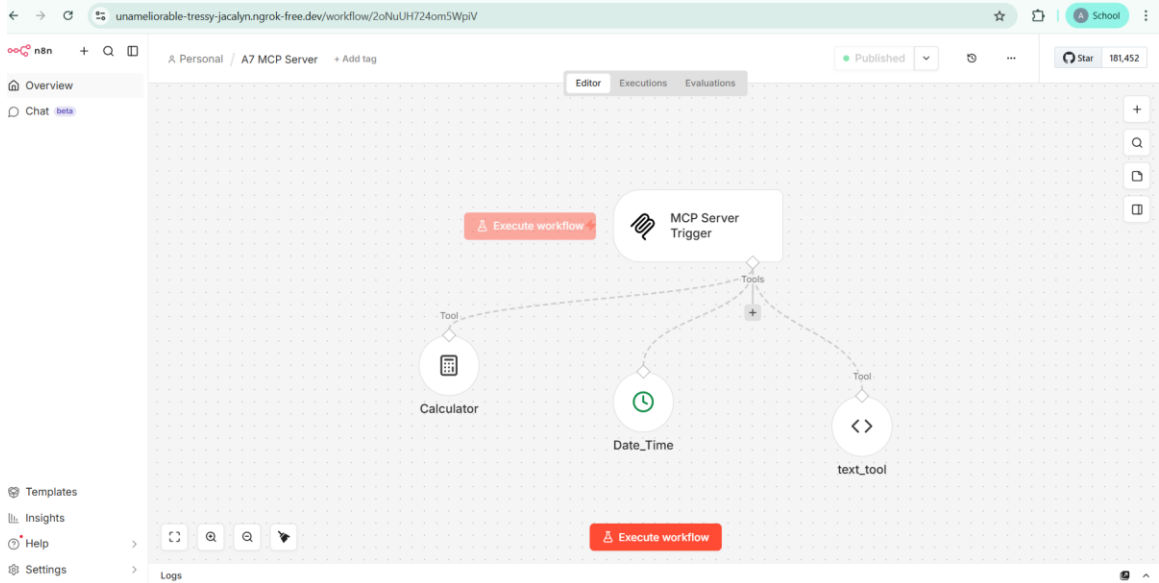
<https://unameliorable-tressy-jacalyn.ngrok-free.dev>

```
ngrok (Ctrl+C to quit)
* One gateway for every AI model. Available in early access *now*: https://ngrok.com/r/ai

Session Status      online
Account             Aashutosh Raut (Plan: Free)
Version             3.37.2
Region              Asia Pacific (ap)
Web Interface        http://127.0.0.1:4040
Forwarding           https://unameliorable-tressy-jacalyn.ngrok-free.dev -> http://localhost:5678

Connections          ttl    opn    rt1    rt5    p50    p90
                     0      0      0.00   0.00   0.00   0.00
```

Figure 4: Public access to the n8n instance using ngrok URL, demonstrating external accessibility.



← → ↺ unameliorable-tressy-jacalyn.ngrok-free.dev/home/workflows

n8n

Overview

Chat beta

Templates

Insights

Help

Settings

Overview

All the workflows, credentials and data tables you have access to

Prod. executions82

Failed prod. executions0

Failure rate0%

Time saved--

Run time (avg.)0s

Workflows

Credentials

Executions

Variables

Data tables

Search

Sort by last updated

A7 MCP Client

Last updated 4 minutes ago | Created 27 March

Personal Published

A7 MCP Server

Last updated 23 hours ago | Created 26 March

Personal Published

My workflow

Last updated 2 days ago | Created 26 March

Personal

Total 3

1

50/page

← → ↺ unameliorable-tressy-jacalyn.ngrok-free.dev/workflow/4a1hECx3pOLv33G/executions/361

n8n

Overview

Chat beta

Templates

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Personal / A7 MCP Client

0 / 3

Published

181,452

Editor

Executions

Evaluations

Executions

Auto refresh

Mar 28, 22:20:47 Succeeded in 386ms

Mar 28, 22:20:38 Succeeded in 849ms

Mar 28, 22:20:27 Succeeded in 847ms

Mar 28, 22:20:19 Succeeded in 708ms

Mar 28, 22:20:12 Succeeded in 389ms

Mar 28, 22:20:05 Succeeded in 450ms

Mar 28, 22:11:53 Succeeded in 382ms

Mar 28, 22:11:45 Succeeded in 78ms

Mar 28, 22:11:36 Succeeded in 749ms

Which executions is this workflow saving?

Mar 28, 22:20:47 Succeeded in 386ms | ID#381

Copy to editor

When chat message received

1 item

AI Agent

Chat Model*

Memory

Tool

Groq Chat Model

Simple Memory

MCP Client

Chat

Logs

AI Agent Success in 382ms 1,032 Tokens

Input

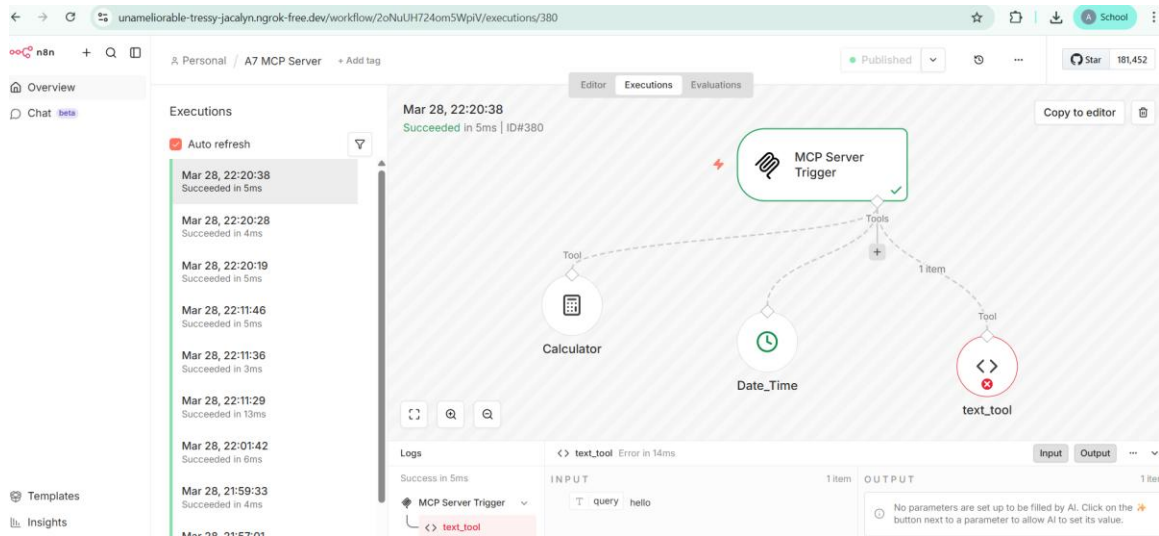
Output

1 item

output

I am using 3 tools: Calculator, Date_Time, and text_tool.

Type message, or



These results confirm that the AI Agent correctly interacts with MCP tools and produces accurate outputs.

Examples of successful executions include:

- Arithmetic computation ($2 + 2 \rightarrow 4$)
- Date retrieval (current date and time)
- Text transformation (HELLO $\rightarrow$ HELLO in uppercase)

The AI Agent correctly identified when to use tools and executed them via the MCP Client. The results confirm that the MCP architecture enables seamless interaction between AI and external tools.

6. Conclusion

The MCP system was successfully implemented using n8n, enabling dynamic tool invocation through an AI Agent. The integration of MCP Server and Client demonstrates how external tools can be accessed and utilized efficiently. Public access via ngrok further validates real-world deployment capability.

Task 2

Overview

Task 2 focuses on integrating the MCP-based AI system with Telegram to create a real-world scheduling assistant. The objective is to allow users to interact with the AI Agent through a messaging interface and perform practical tasks such as creating calendar events.

This task demonstrates how natural language input from users can be processed by an AI Agent and converted into structured actions using external tools, specifically Google Calendar.

Implementation

The Telegram integration was implemented using an n8n workflow consisting of the following components:

- Telegram Trigger: Receives user messages
- Edit Fields: Processes incoming message data
- AI Agent: Interprets user intent and determines required actions
- Groq Chat Model: Provides natural language understanding
- Simple Memory: Maintains conversation context
- Google Calendar Nodes:
 - Create Event: Creates calendar events
 - Get Events: Retrieves scheduled events
- Telegram Send Message: Sends responses back to the user

When a user sends a scheduling request (e.g., creating a project schedule), the AI Agent processes the request and generates structured event details. It then invokes the Google Calendar tool to create events with specified dates, times, and titles.

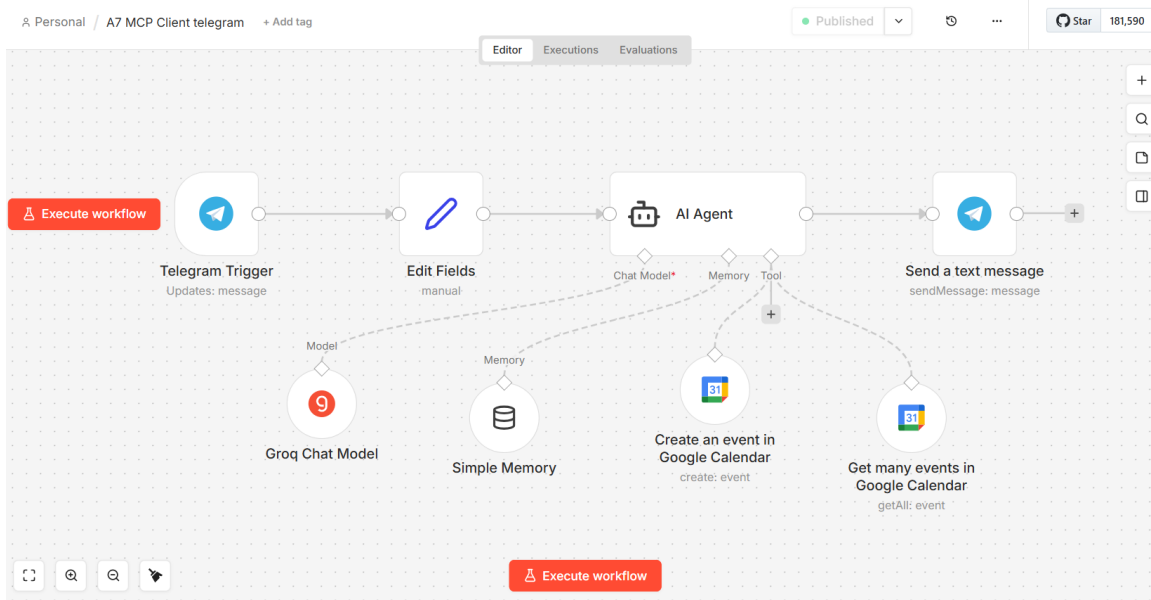
The system also allows users to query their schedule, demonstrating bidirectional interaction between the user and the AI system.

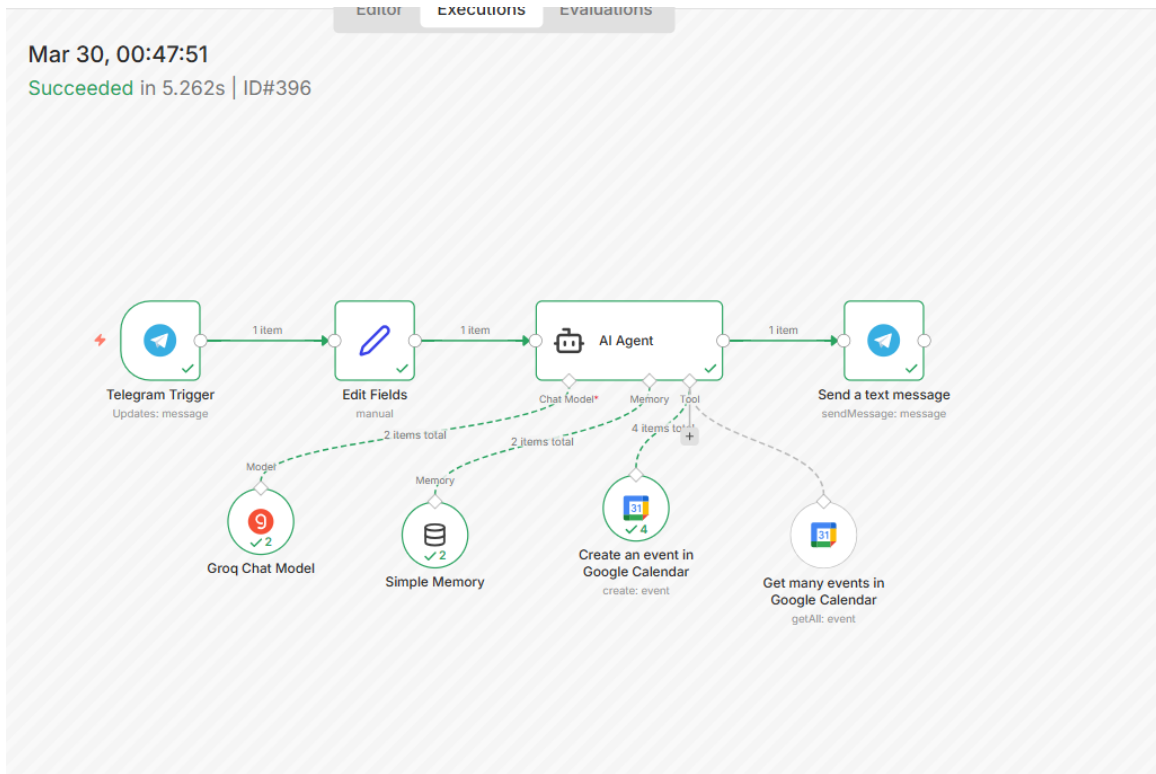
The Telegram bot acts as a user interface for the MCP-powered system.

Workflow:

1. The user sends a message via Telegram.
2. The Telegram Trigger node receives the message in n8n.
3. The message is passed to the AI Agent.
4. The AI Agent processes the request and determines whether to use tools.
5. For scheduling tasks, the AI Agent invokes Google Calendar tools.
6. Events are created, and the result is sent back to the user via Telegram.

This integration demonstrates how MCP-based systems can be connected to real-world applications.





Overview Chat **Beta**

A Personal / A7 MCP Client telegram + Add tag Published 181,590

Executions

Auto refresh

- Mar 30, 00:22:50 Succeeded in 4.611s
- Mar 30, 00:17:54 Error in 512ms
- Mar 30, 00:07:00 Error in 457ms
- Mar 30, 00:05:37 Succeeded in 2.411s
- Mar 29, 23:59:50 Error in 50ms
- Mar 29, 23:44:32 Error in 20ms
- Mar 29, 23:41:59 Error in 27ms
- Mar 29, 23:34:32 Error in 14ms
- Mar 29, 23:33:31 Error in 9ms

Which executions is this workflow saving?

Editor Executions Evaluations

Mar 30, 00:22:50 Succeeded in 5.262s | ID#392

Telegram Trigger Updates: message

Edit Fields manual

AI Agent

Send a text message sendMessage: message

Groq Chat Model

Simple Memory

Create an event in Google Calendar create: event

Get many events in Google Calendar getAll: event

Logs

Success in 4... 2,753 Tok...

AI Agent Success in 514ms 2,753 Tokens

INPUT

- # update_id 622589168
- # message
- # message_id 11
- # from
- # id 8360196653
- is_bot false
- first_name Mr Jones
- language_code en
- chat
- # id 8360196653
- first_name Mr Jones

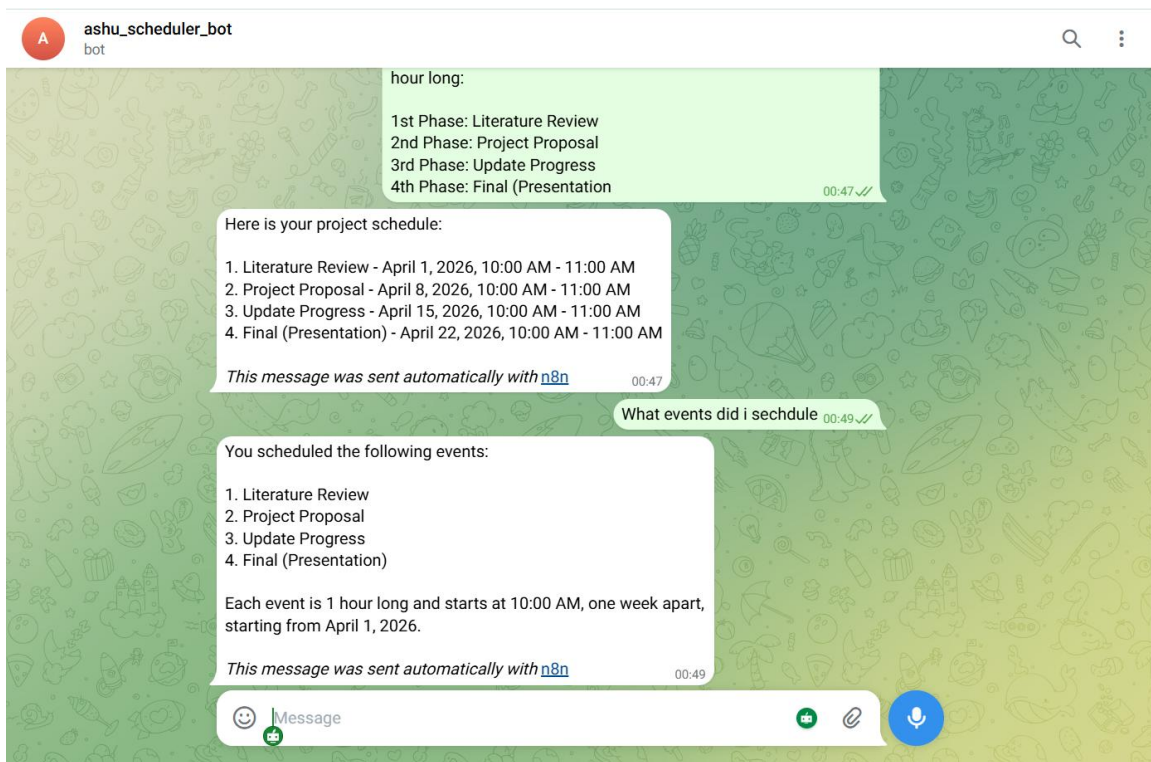
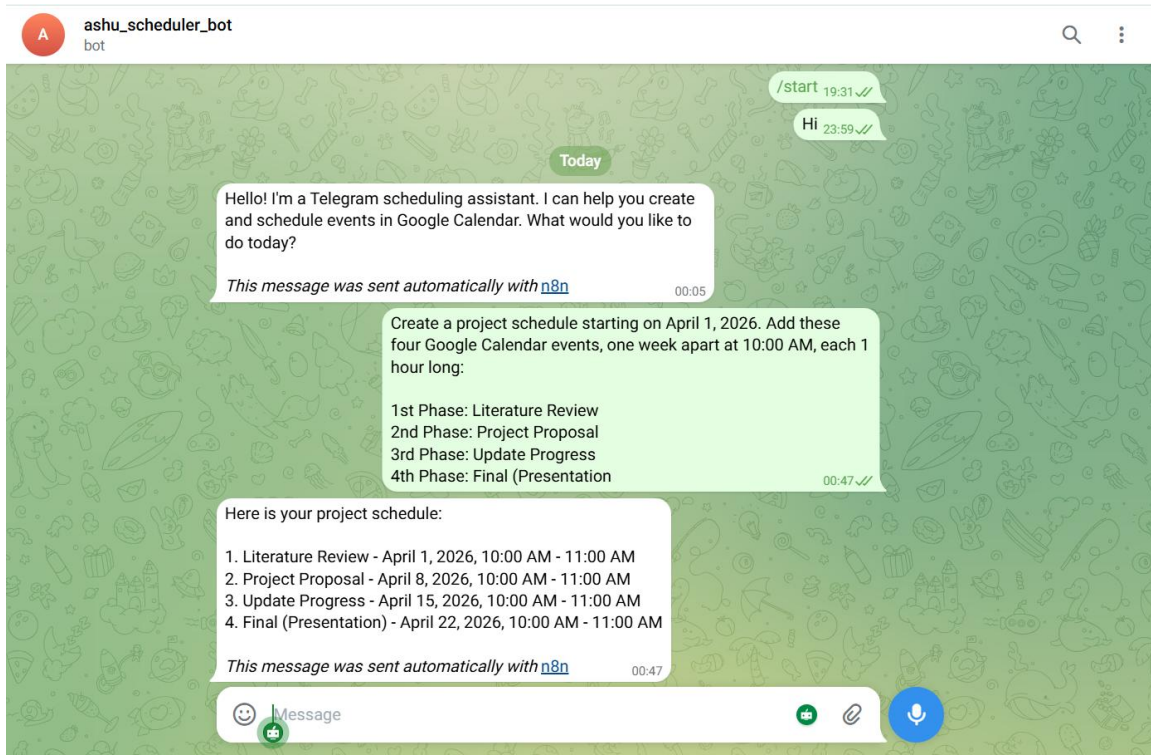
OUTPUT

1 item

output

Here is your project schedule:\n\n1. Literature Review - April 1, 2026, 10:00 AM - 11:00 AM\n2. Project Proposal - April 8, 2026, 10:00 AM - 11:00 AM\n3. Update Progress - April 15, 2026, 10:00 AM - 11:00 AM\n4. Final Presentation - April 22, 2026, 10:00 AM - 11:00 AM

Telegram Conversation



Google api for calander for n8n

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Dismiss [Start free](#)

Google Cloud

n8n-calendar

Search (/) for resources, docs, products, and more

Search

Google Auth Platform / Clients / Client: 678464569527-r3jp5s4tklqkxm34pgrutsgdedr34hp.apps.googleusercontent.com

Overview

Branding

Audience

Clients

Data Access

Verification Center

Settings

Client ID for Web application

Delete

Name *
n8n-calendar

The name of your OAuth 2.0 client. This name is only used to identify the client in the console and will not be shown to end users.

The domains of the URIs you add below will be automatically added to your [OAuth consent screen](#) as [authorized domains](#).

Authorized JavaScript origins

For use with requests from a browser

+ Add URI

Authorized redirect URIs

For use with requests from a web server

URIs: 1 *
http://localhost:5678/rest/oauth2-credential/callback

Additional information

Client ID

678464569527-r3jp5s4tklqkxm34pgrutsgdedr34hp.apps.googleusercontent.com

Creation date

March 29, 2026, 10:40:40 PM GMT+7

Last used date

March 29, 2026 (Note: this data could be delayed by a day or more.)

Inactive OAuth clients are subject to deletion if they are not used for 6 months. You will be notified of deletion due to inactivity, and can restore clients up to 30 days after deletion. [Learn more](#)

Client secrets

If you are in the process of changing client secrets, you can manually rotate them without downtime. [Learn more](#)

Client secret

****ug0W

Download

Creation date

March 29, 2026, 10:40:40 PM GMT+7

Status

Enabled

Google Calendar — Project Phases

Calendar

Today

<

>

April 2026

Search

Help

Settings

Month

Calendar

Checklist

Grid

AIT

A

+ Create

April 2026

S M T W T F S

29 30 31 1 2 3 4

5 6 7 8 9 10 11

12 13 14 15 16 17 18

19 20 21 22 23 24 25

26 27 28 29 30 1 2

3 4 5 6 7 8 9

Meet with...

Search for people

Booking pages

30 min with Aashuto...

Time Insights

APR 1 - 30, 2026

0 hr in meetings

More insights

My calendars

SUN 29

MON 30

TUE 31

WED Apr 1

THU 2

FRI 3

SAT 4

5

6 Chakri Day

7

8 9pm 1st Phase: Literature

9

10

11

12

13 Songkran

14 Songkran Holiday

15 Songkran Holiday

16

17

18

19

20

21

22 9pm 2nd Phase: Project F

23

24

25

26

27

28

29 9pm 3rd Phase: Update P

30

24

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28

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30

May 1 Labor Day

2

Result Summary

The Telegram-based scheduling assistant successfully performed the following tasks:

- Interpreted natural language scheduling requests
- Created multiple calendar events with correct dates and times
- Assigned appropriate titles to events
- Retrieved and displayed scheduled events upon user request

For example, the system created a four-phase project schedule with events spaced one week apart at 10:00 AM, each lasting one hour.

The results confirm that the AI Agent can effectively translate user intent into structured calendar operations.

Event	Date	Time	Description
1st Phase: Literature Review	April 1, 2026	10:00–11:00	Review related papers
2nd Phase: Project Proposal	April 8, 2026	10:00–11:00	Prepare proposal
3rd Phase: Update Progress	April 15, 2026	10:00–11:00	Present progress
4th Phase: Final Presentation	April 22, 2026	10:00–11:00	Final demo

Conclusion

The Telegram integration successfully demonstrates a real-world application of the MCP-based AI system. By combining natural language processing with tool-based automation, the system enables users to manage their schedules efficiently through a conversational interface.

This implementation highlights the potential of AI-driven automation in everyday applications and showcases how MCP can be extended beyond basic tool usage to practical, user-facing systems.